

Teaching Statement

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Teaching philosophy. I have taught mathematics in many settings: coaching olympiad students in South Africa, supervising first-year undergraduates at Warwick, and lecturing a Master's course in algebraic number theory at Charles University. Across all of these, my aim has been the same: to help students move from following mathematical arguments to constructing their own. I encourage students to ask questions, explain their reasoning, and treat uncertainty and mistakes as a normal part of doing mathematics. I also try to make sure that a class is not dominated by its most confident participants. Some of the most useful discussions begin with a seemingly simple question that reveals a difficulty several students share.

Explicit examples are central to how I teach. Many ideas in number theory only show their full power at a high level of abstraction. Students engage with that abstraction far more readily once they have seen a concrete problem that it solves. I therefore try to use examples to motivate the theory, not only to illustrate it after the fact.

Lecturing: Advanced Algebraic Number Theory. I am the instructor for the Master's-level course *NMAL431 Advanced Algebraic Number Theory* at Charles University.¹ I taught it in the winter semester of 2025/26 and am teaching it again in 2026/27. The course covers Dedekind domains, localisation and completion, the Galois theory of local and global fields, and an introduction to class field theory. I give the weekly lectures and fortnightly exercise classes, prepare exercise and homework sheets, and conduct the oral examinations. Teaching the course has given me responsibility for the complete structure of an advanced mathematics course, from deciding how material should be organised and motivated to designing exercises and assessing students' understanding.

The course ends with the main statements of global class field theory. This is where abstraction is hardest to motivate, so I close with a classical application: for suitable n , an odd prime p is of the form $x^2 + ny^2$ if and only if p splits completely in the Hilbert class field of $\mathbb{Q}(\sqrt{-n})$. A question about which primes a simple quadratic form represents turns out to be governed by the arithmetic of an abelian extension of an imaginary quadratic field. This result lets students see what the machinery of the course is for, and it connects the course back to elementary number theory they already know.

The classes are small (seven students in the first year and five in the second), which lets me adapt the pace to the students in the room. It also lets me make the oral examinations genuinely interactive. Students present a proof from the course, but I also ask them why particular steps are needed, what goes wrong if a hypothesis is dropped, and how the argument relates to other parts of the course. These questions separate a student who has memorised a proof from one who understands it.

Teaching at Warwick. Before moving to Charles University, I spent four years as a Senior Graduate Teaching Assistant during my PhD at the University of Warwick. I led support classes for modules ranging from introductory first-year classes to advanced third- and fourth-year modules, including Algebraic Number Theory, Modular Forms, Ring Theory, Groups and Representations, Combinatorial Optimisation, and Maths by Computer. My responsibilities included weekly support classes, computational laboratory sessions, marking assignments, and invigilating and marking final examinations.

For three years I also supervised small groups of first-year mathematics students twice each week. These supervisions covered all of the core first-year subjects, including real analysis, abstract algebra, differential equations, linear algebra, and foundations of mathematics. Working with the same students throughout the year let me watch their development closely. I could

¹Course webpage: <https://rvisser7.github.io/nmal431/>

identify recurring difficulties, give detailed feedback on written work, and shift the emphasis of each session to what individual students needed. This experience shaped my teaching more than any other. For new undergraduates, the hardest step is often learning how to think about a problem when no worked example applies, rather than mastering any particular theorem. So rather than showing students a solution, I would ask them where exactly they were stuck and which definitions or results might be relevant, and then let them take the next step themselves.

Supervision and mentoring. More recently, I have also had the opportunity to supervise students individually. At Charles University I supervised a Bachelor's thesis on cohomological aspects of class field theory. Working through Milne's notes, the student and I built up the cohomological machinery and gave proofs of the main theorems of class field theory. This is ambitious material for a Bachelor's student. He defended the thesis successfully in June 2026 and received the highest grade (1, excellent). I also supervised a summer project on rational points on elliptic curves, based on Silverman and Tate's book. Alongside the theory, the student carried out explicit computations in SageMath. Seeing the group law, torsion points and rational points computed on actual curves made the theory tangible in a way that reading alone does not. In 2026 I also served as the opponent (external examiner) for a Bachelor's thesis, writing the evaluation report and examining the student at the defence.

I particularly enjoy supervision because it lies close to the boundary between teaching and research. The goal is no longer simply to guide a student through a fixed syllabus, but to help them learn how to read mathematics independently, decide which questions to investigate, recognise gaps in an argument, and present complicated ideas clearly.

Olympiads and outreach. My teaching and mentoring experience also extends beyond formal university courses. I have been involved for many years in mathematics and programming competitions, first as a participant and subsequently as a coach, team leader, and organiser. I have coached at South African national mathematics camps, helped set and mark tests and assignments used in the selection of olympiad teams, led university teams at the International Mathematics Competition, and served as a team guide at the International Mathematical Olympiad. More recently, I co-organised the Department of Algebra's autumn school at Charles University, where I reviewed students' proposed talks and abstracts and advised them on preparing mathematical presentations.

Future teaching. I would be glad to contribute broadly to undergraduate and graduate teaching, including core courses such as calculus and linear algebra, as well as courses in algebra and number theory. My research in computational number theory gives me a natural way to connect abstract theory with explicit computation, and I would like to build this into my teaching. I would also like to develop an advanced course on elliptic curves or computational number theory, in which students test conjectures and discover patterns with SageMath or PARI/GP before proving them. Such a course could draw on resources like the LMFDB, to which I contribute, and would naturally generate accessible student projects.

Whatever the course, I want students to leave with more than a collection of results and techniques. I want them to leave able to approach new mathematics with confidence, ask productive questions, and keep learning on their own.